



UNIVERSITY OF
BIRMINGHAM

Superconductivity

Lecture 05 – “Geometry Matters”

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Previously, heat capacity ...

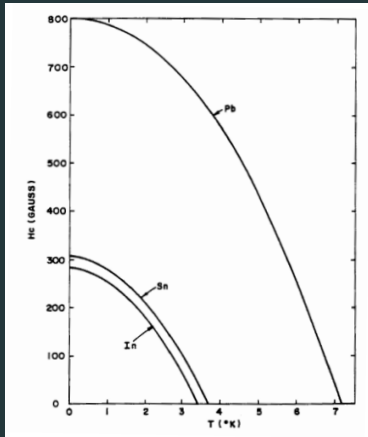
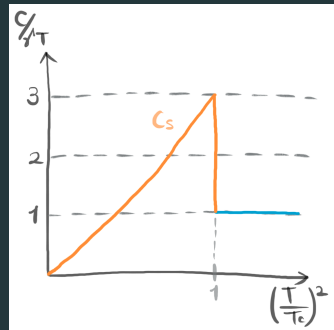


Figure 1: $H - T$ phase diagram of elemental metals, Pb, Sn and In.

Using

$$H_c(T) = H_c(0) \left[1 - \left(\frac{T}{T_c} \right)^2 \right]$$

and $\chi = -1$, one can predict



Quantitative agreement

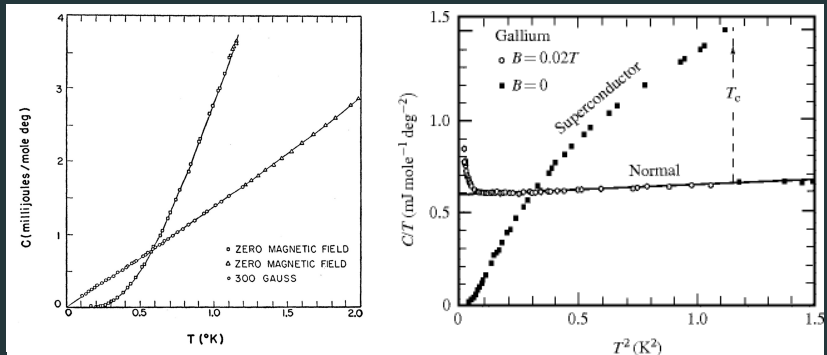


Figure 2: Temperature dependence of specific heat of Al (left, N. E. Phillips, Phys. Rev. B **114**, 676, 1959) and Ga (right, C. Kittel, Introduction to Solid State Physics, John Wiley & Sons, 1996) in zero and finite fields.

Previously, critical currents

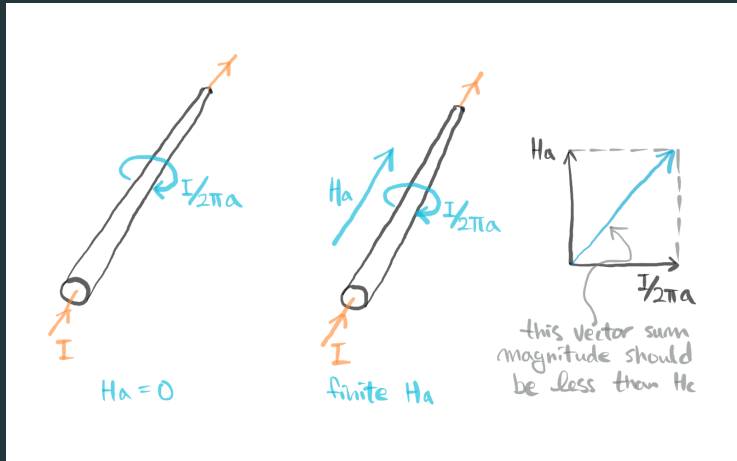


Figure 3: Currents and fields for a superconducting wire.

Critical currents and fields

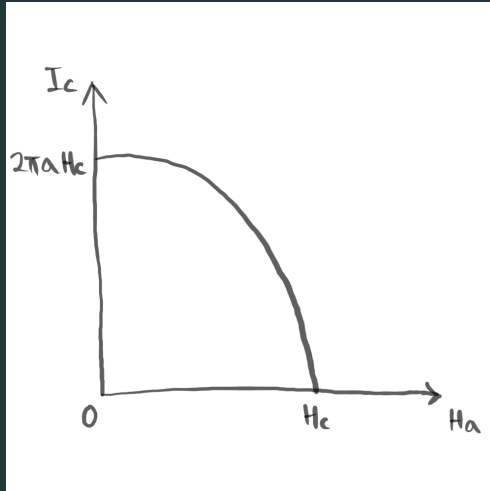


Figure 4: Critical current vs critical field diagram.

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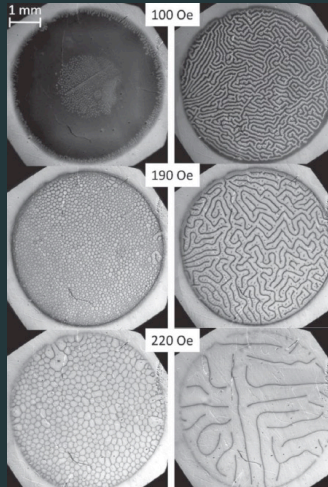


Figure 5: Magneto-optical images of a Pb crystal at 5 K.

Demagnetisation

We have neglected one practical aspect of magnetism, that is, demagnetisation (cf. depolarisation in dielectric materials).

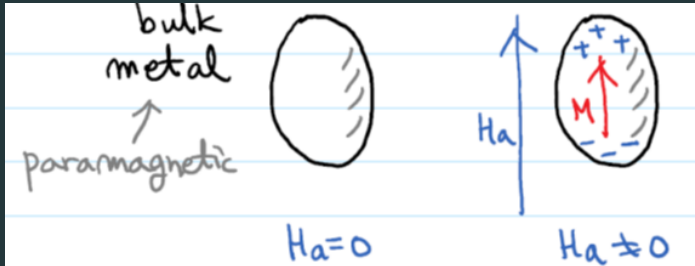


Figure 6: Materials' response to a magnetic field.

$$H_i = H_a - H'. \quad (1)$$

Demagnetisation depends on 'geometry'

Sample geometry matters:

- an infinitely long rod $\rightarrow H' = 0$,
- a very thin sheet $\rightarrow H'$ largest.

$M(H_a)$ matters since demagnetising $H' \propto M$:

- Small effects for metals $\leftarrow M \simeq 0$ since $\chi_P \simeq 0$
- Large effects for ferromagnets or superconductors, i.e., H' largest.

Ellipsoid

Moreover, $\mathbf{H}'$ is position dependent for a sample of arbitrary shape
→ all too much complicated...

However, it can be made uniform throughout if we use a special shape, an ellipsoid.

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1, \quad (2)$$

where a , b and c are dimensions along the principal axes.

This could be useful as

- an infinite rod: $c \rightarrow \infty$,
- a very thin plate: $a, b \rightarrow \infty$,

Demagnetising field for an ellipsoid

The original definition using vectors

$$\mathbf{H}_i = \mathbf{H}_a - \mathbf{H}'. \quad (3)$$

reduces to a scalar expression

$$H_i = H_a - nM, \quad (4)$$

where the demagnetisation factor $0 \leq n \leq 1$.

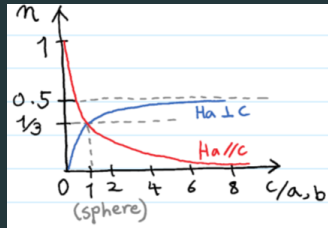


Figure 7: Demagnetisation factor of an ellipsoid as a function of the ratio of the principal axes lengths.

Demagnetisation in superconductors

'De'magnetisation in superconductors (1/2)

Something special with demagnetising field in superconductors:
It ADDS UP to the applied field, thereby increasing the total effective field, unlike a para- or ferro-magnet.

$$\begin{aligned}H_i &= H_a - nM \\ &= H_a + nH_i,\end{aligned}$$

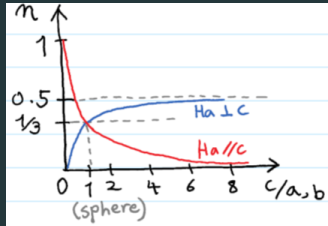
owing to the Meissner effect $M = -H_i$. By rearranging, one obtains

$$H_i = \frac{1}{1-n} H_a,$$

that is, $H_i \geq H_a$ as $0 \leq n \leq 1$

'De'magnetisation in superconductors (2/2)

$$H_i = \frac{1}{1-n} H_a.$$



- Sphere $\rightarrow H_i = 1.5H_a$
- A long rod in a transverse field $\rightarrow H_i = 2H_a$

Critical field can be reached in a lower applied field ... ?

Coexistence of normal & superconducting regions

Dilemma

What happens when the internal field H_i has reached the critical field H_c while H_a remains much less than H_c ?

- As soon as the superconducting state becomes normal metallic, the material loses its perfect diamagnetism and becomes a Pauli paramagnet such that $H_i \simeq H_a$.
- Meanwhile, since the applied field $H_a = (1 - n)H_c$ is still smaller than H_c , one is led to the situation of $H_i < H_c$.
- This means that the internal field has not yet reached to the critical value, thus superconductivity should have persisted.
- Of course, this again restores perfect diamagnetism such that the internal field reaches the critical value destroying again superconductivity.

So, we are stuck ...

Relieve frustration through coexistence

For $H'_c \leq H_a \leq H_c$, superconducting and normal metallic states coexist in patches

→ Note that this introduces 'boundaries' or 'interfaces' between the two states.

→ What are the configuration of those boundaries or domains?

Recall the boundary conditions from electromagnetism:

1. $B_{\perp}$ be continuous across the interface,
2. $H_{\parallel}$ be continuous across the interface.

Boundary conditions

1. $B_{\perp}$ be continuous across the interface:
Since $B = 0$ inside a superconductor, the interface should align parallel to the applied field such that $B_{\perp}$ is zero on the metallic side.
2. $H_{\parallel}$ be continuous across the interface:
At the interface, $H_i = H_c$ to maintain equilibrium, since otherwise one region would grow over the other.
 - if $H_i > H_c$, having more of normal states would gain energy,
 - if $H_i < H_c$, superconducting states would be favoured.

Thus, at the very interface and throughout the sample, $H_i = H_c$.

Intermediate states

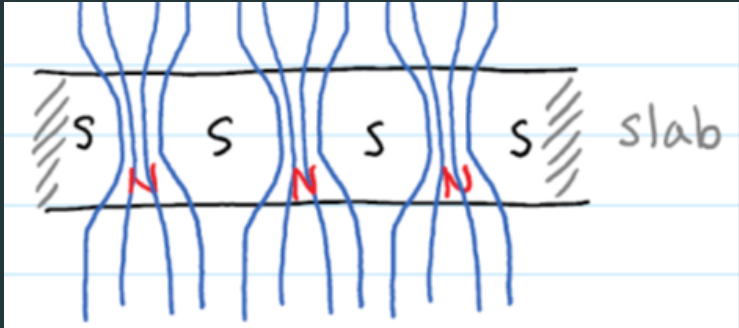


Figure 8: Schematic drawing of magnetic fields inside a superconducting slab.

Imaging domains

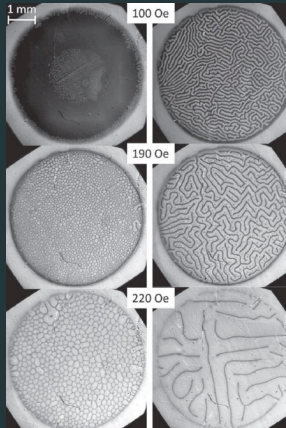


Figure 9: Magneto-optical images (using Faraday rotation) of an intermediate state of a Pb single crystal at 5 K in different magnetic fields [R. Prozorov, Phys. Rev. Lett., 98, 257001 (2007)]. Bright regions correspond to the normal state and dark regions to the superconducting state

Interface energy

Interface energy

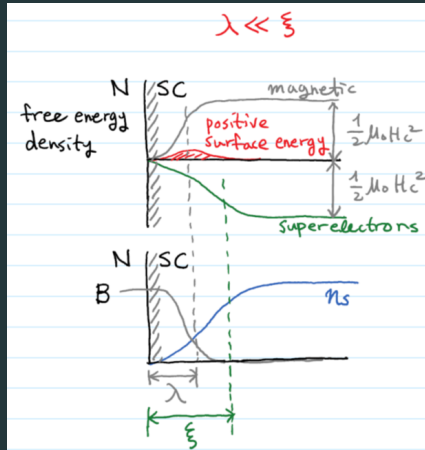


Figure 10: Magnetic flux and superconducting electron density profile across the normal metal-superconductor boundary